Loan Approval Prediction Agent - Project Report

1. Objective:

To predict if a loan will be approved or rejected using customer data.

2. Workflow:

- Dataset: Loan data (14 original columns, 9000+ rows)
- Preprocessing: Converted categorical columns (Gender, Education, Home Ownership, Loan Intent) using One-Hot Encoding. Total features increased from 14 to 22.
- Cleaning: Dropped 2 low-importance features after analysis, leaving 20 final features.
- Model: RandomForestClassifier with 80/20 train-test split (Train: 36000, Test: 9000).

3. Results:

- Accuracy: 92.35%
- Precision: 0.94 (Class 0), 0.86 (Class 1)
- Recall: 0.96 (Class 0), 0.78 (Class 1)
- The model correctly predicts 92 out of 100 loans.
- Key drivers: Credit Score, Credit History, and Person Income vs Loan Amount.

4. Conclusion:

The high accuracy shows loan approval is highly dependent on credit behavior. This model can be deployed as an AI Agent/API to automate loan decisions and reduce manual review time. Future work is to convert this model into a conversational loan agent like Meta AI for instant customer interaction.

Model Performance:

- Train size: 80%
- Test size: 20%
- Algorithm: Random Forest
- Final Accuracy: 0.9235